

Skills Summary

Solving Systems by Substitution

Use this reference while completing your worksheet or when studying.

Skill 1 — When one equation is already solved

The move: if one equation says $y = (\text{something in } x)$, replace y in the *other* equation by that something. The result is one equation in one variable. **Two habits that prevent most errors:** wrap the substituted expression in brackets, and put it into the equation it did *not* come from — substituting back into its own equation only ever gives $0 = 0$.

Example: Solve $y = 3x + 1$ and $2x + y = 11$.

Step 1. The first equation already hands me y as an expression in x , so substitution turns the second equation into one I can solve straight away.

Step 2. $2x + (3x + 1) = 11$, so $5x + 1 = 11$, $5x = 10$, $x = 2$.

Step 3. Back-substitute: $y = 3(2) + 1 = 7$. Check in the second equation: $4 + 7 = 11$.

$\Rightarrow (2, 7)$

Try it: Solve $y = 2x + 1$ and $x + y = 10$.

check: $(2, 8)$

Skill 2 — Isolating a variable first

When neither equation is ready: pick the variable whose coefficient is 1 or -1 and isolate it. That choice is what keeps fractions out of the substitution. Isolating a variable is itself a legal move — add or subtract the same amount on both sides, then divide — so the new equation still says exactly what the old one said.

Example: Solve $x + 2y = 11$ and $3x - y = 5$.

Step 1. Nothing is solved for a variable yet, so I isolate the one with coefficient 1 — the x in the first equation — to avoid fractions.

Step 2. $x = 11 - 2y$. Substituting into the second: $3(11 - 2y) - y = 5$.

Step 3. $33 - 6y - y = 5$, so $-7y = -28$ and $y = 4$; then $x = 11 - 8 = 3$.

$\Rightarrow (3, 4)$

Try it: Solve $x + 3y = 13$ and $2x - y = 5$.

check: $(4, 3)$

Skill 3 — When the variable disappears

Two endings that look like failures and are not: sometimes the substitution cancels every x and leaves a statement with no variable in it.

- A *false* statement such as $-4 = 0$: no solution. The lines are parallel.
- A *true* statement such as $8 = 8$: infinitely many solutions. The two equations describe the same line.

Write which of the two it is; do not try to solve the leftover statement.

Example: Solve $y = 2x + 5$ and $4x - 2y + 6 = 0$.

Step 1. The first equation is ready to substitute, so I do that and watch what happens to the variable rather than assuming a number will come out.

Step 2. $4x - 2(2x + 5) + 6 = 0$, so $4x - 4x - 10 + 6 = 0$, which is $-4 = 0$.

Step 3. Every x has cancelled and the statement left behind is false, so nothing can satisfy both.

$\Rightarrow$ no solution

Try it: Solve $y = 3x + 2$ and $6x - 2y + 1 = 0$.

check: no solution